TMD-364



Specifications	
MCU	ARM Cortex-M23 @48MHz
OS	RTOS/Custom Scheduler
GSM	4G CAT1 Module 4G Bands : B1, B2, B3, B4, B5, B7, B8, B20, B28, B38, B40, B41, B66 2G Bands : 850/900/1800/1900 MHz
GNSS	• GPS (L1), GLONASS (L1, FDMA), Galileo (E1), Beidou (B1) • IRNSS(AIS-140 Variant) • Position Accuracy 1.5m, • Sensitivity: -162dbm • Time To First Fix (@ -130 dBm) - Hot Start: 1s - Cold Start: 34s
Bluetooth(Optional)	Bluetooth Low Energy (BLE) 5.2 upto 2Mbps
SIM	• Embedded SIM or Nano SIM CARD
Antennas	Internal GPS, GSM, BLE
Interface	1 – RS232/RS-485, 1-CAN
I/O	Analog inputs – 2, Digital inputs – 4, Digital outputs - 2
Storage	SPI Flash up to 128Mb
Motion Sensor	6-axis IMU (accelerometer + Gyroscope)
Protocols	TCP/HTTP
LED Indicator	GPS and GSM
Built-in Battery	1000mAH (Up to 4 Hours backup)
Connector	MX23 Series 18-Pin connector from JAE
Power Supply	8 – 75V
Operating Temperature	-20 to +60 °C with battery -40 to +80 °C without battery
Humidity	95% R.H. at 50°C
Dimension	129 x 75 x 34 mm
Weight	150 grams
IP Rating	IP67

Features

- Real time Vehicle Tracking along with ability to log 40,000 records.
- Remote vehicle health monitoring with programmable update frequency over Cellular Data/SMS
- Vehicle Diagnostics
- Firmware Over The Air(FOTA)
- Configuration over SERIAL/DATA/SMS
- Lower power Consumption and Power optimization using power modes for Extended battery support
- Internal Battery Backup
- Device Alerts
    - Geo-Fencing
    - Over Speed
    - Harsh Breaking/Harsh Acceleration,
    - Fatigue
    - Towing
    - Tampering
    - Ignition
    - Power Failure

Pinout

Pin No	Pin Name	Description	Remarks
1	Digital_IN_4	DIGITAL INPUT 4	UPTO 45V
2	GROUND	POWER GROUND	CONNECTR BATTERY -VE
3	RS232-RX	RS-232 RX	
4	RS232-TX	RS-232 TX	
5	DIGITAL_IN_2	DIGITAL INPUT 2	UPTO 45V
6	DIGITAL_IN_3	DIGITAL INPUT 3	UPTO 45V
7	GROUND	GROUND FOR CAN BUS	
8	CAN_H_1	CAN HIGH	
9	CAN_L_1	CAN LOW	
10	VIN	POWER SUPPLY INPUT	UPTO 75V
11	ANALOG_IN 1 / IGNITION	ANALOG INPUT 1 or IGNITION	UPTO 45V
12	ANALOG_IN 2	ANALOG INPUT 2	UPTO 45V
13	DIGITAL_IN_1	DIGITAL INPUT 1	UPTO 45V
14	DIGI_OUT_1	DIGITAL OUTPUT 1	CURRENT SINK 1A@45V
15	DIGI_OUT_2	DIGITAL OUTPUT 2	CURRENT SINK 1A@45V
16	GROUND	POWER GROUND	CONNECTR BATTERY -VE
17	RESERVED		
18	RESERVED		

Compliance

EMI\EMC	Radiate Emissions (RE)	CISPR 25:2005
	Conducted Emissions (CE)	CISPR 25:2008
	Radiated Immunity (RI)	ISO 11452 - 2 : 2004
	Conducted Immunity (CI)	ISO 11452 - 4 : 2003
	Pulse Test Pulse 1, 2a, 2b, 3a & 3b, 4, 5a and 5b	ISO 7637 - 2 : 2011
	Transient Emission	ISO 7637 - 2 : 2011
Electrical	Minimum Supply Voltage	ISO 16750-2: 2006
	Maximum Supply Voltage	
	Supply imposed alternating voltage	
	Slow decrease and increase of supply voltage	
	Momentary drop in supply Voltage	
	Rest behavior of voltage drop	
	starting profile test	
	Reverse Voltage	
	Load dump	
	Short circuit protection signal circuits	
	Electrostatic Discharge Test	
Climatic	Thermal cyclic Thermal shock High Temperature and Low Temperature	ISO 16750-4
	Humidity Test	
Mechanical	Vibration test	ISO 16750-3
	Mechanical shock test	
	Drop test	